

TITLE

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Abstract

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Chapter 1

CONSTRUCTING TILING SPACES

We have different methods of constructing tiling spaces. The first is to be generated by a set of tilings. If we take $\mathbb{R}^n$ and subdivide it into different pieces, then each individual piece is called a *tile*, where each tile overlaps another only at its border. The union of all the tiles is called a *tiling*. For our examples we will only have a finite number of tile types, up to translation, called prototiles. Each prototile is a polytope. For a simple example let $n = 1$, and let $T = \cup_{i \in \mathbb{Z}} t_i$, where $t_i = [i, i + 1]$. Then each tile only intersects another tile at its boundary points and $T = \cup t_i = \mathbb{R}$. A *translation* of a tiling T is a shift of the tiling by some vector $x \in \mathbb{R}^n$. Let T be as above, then $T - 1$ is a translation of T to the right by one unit.

We can place a metric on tilings by considering two tilings close (within ϵ) if after a small translation (ϵ) these tiles agree on a large radius ($1/\epsilon$) about the origin. It is difficult to state the metric formally. In the following definition, the closure and therefore, the topology, is given by this metric.

Definition 1 *A tiling space Ω_T is a set of tilings that has the following two properties*

- 1) *For each $T \in \Omega_T$ and each $x \in \mathbb{R}^n$ implies $T - x \in \Omega_T$*
- 2) *$\overline{\{T - x \mid x \in \mathbb{R}^n\}} \subseteq \Omega_T$ for each $T \in \Omega_T$*

Chapter 2

INTRODUCTION

2.1 daf

2.1.1 asdf

$$\sum_{i=0}^n i = \frac{n(n+1)}{2} O(n^2)$$

In this paper we are interested in determining if two aperiodic tilings are essentially different (combinatorially). Here, aperiodic means no translation of the tiling produces the original tiling. Aperiodic tilings have a long history. For a more complete account see Sadun's *Topology of Tiling Spaces* [?] or Radin's *Miles of Tiles* [?]. In the 1960's, Hao Wang [?] studied tilings in relation to decidability problems. He proposed the Domino Problem concerning whether the plane could be tiled using Wang Dominoes. Robert Berger [?] showed this problem was undecidable, which immediately implies the existence of an aperiodic tiling. In this proof, he constructed a 20,426 wang tile example. It has been studied by many since then, and in 1974, Roger Penrose [?] reduced the number of prototiles required to only two, with the Kites and Darts tiling, Figure 2.1. Substitutions provide straightforward ways to produce aperiodic tilings. As such, many examples are known. It is an open problem, referred to as the Einstein problem, to construct an periodic

tiling with one prototile. The pinwheel tiling, as suggested by John Conway and constructed by Charles Radin, is constructed with a single tile, in that it uses a single shape, but one must use this tile in infinitely many orientations, along with their reflections.

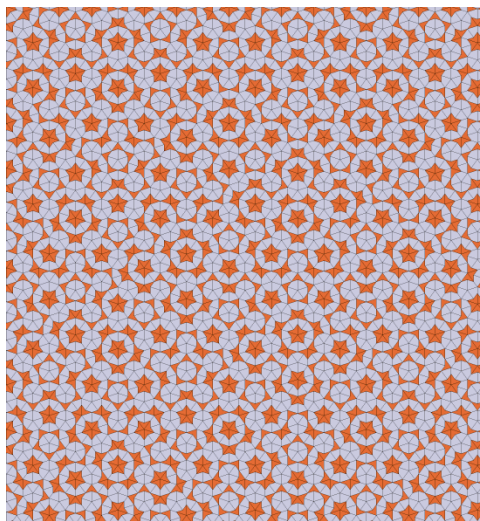


Figure 2.1: Penrose kite and dart tiling from the Tilings Encyclopedia Website (http://tilings.math.uni-bielefeld.de/substitution_rules/)